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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech (CSE)

Semester - I

Examination, Dec 2009-10

EC 701 Advance High Speed Networks

Date: 29/12/2009, Tuesday Time: 1.00 p.m to 4.00 p.m Maximum Marks: 70

Instructions:

1. Write your answers on two separate sheets for each section.
2. Figures to the right are Marks
3. Assume Suitable data and draw the figure if necessary.
4. Attempt all six questions.

Section I

- Q1 a)** Explain each field of 802.11 General frame Structure. 4
b) Discuss Route selection in MPLS 4
c) What is Flapping and Looping? 3
- Q2 a)** Suppose that an 11 Mbps 802.11b LAN is transmitting 64 byte frames back to back over a radio channel with a bit error rate of 10^{-7} . How many frames per second will be damaged on average? 4
b) Discuss the real time and non real time services for ATM in detail 6
c) Compare Cipher Block Chaining with cipher feedback mode in terms of the number of encryption operation to transmit a large file. 2

OR

- Q2 a)** Using the RSA public key cryptosystem with $a=1, b=2$ etc. 4
(a) If $p=7$ and $q=11$ list five legal values for d
(b) If $p=13, q=31$ and $d=7$, find e .
b) With neat sketch discuss the basic architecture for an interface between User and Network. 8
- Q3 a)** Discuss with neat sketch the operation of MPLS in detail. 7
b) Design an IP address scheme using **VLSM CIDR** for given scenario such that minimum number of IP address are wasted: A campus has three subnets They have 47, 28, 12 PCs. respectively. Use class C Network address 200.1.1.0/24. 5

OR

- Q3 a)** Explain time to leave processing in MPLS in detail 8
b) A large number of consecutive IP addresses are available starting at 198.16.0.0. Suppose that four organization A, B, C and D, request 4000, 2000, 4000 and 8000 addresses respectively. Give first and last ip address assigned and its subnet mask in w.x.y.z/s notation. 4

Section II

- Q4) Do as directed**
- a) Compare Distance Vector Routing and Link State Routing. 4
 - c) Define the following terms for MPLS. 4
Forwarding equivalence Class(FEC) 2) Label Switched Path 3) LSR
 - e) Find whether the given IP address is host address, network address or broadcast address. If host address then list their network and broadcast address. 3
1) 192.168.0.108/27 2) 10.10.0.112/28 3) 72.19.255.255/15.
- Q5 a)** Explain Distributed Bellman Ford algorithm. 5
- b) Explain types of AAL in ATM 4
 - c) List out the various standards of Gigabit Ethernet cabling. 3
- OR**
- Q5 a)** Discuss various Link costs in Link state Routing Algorithm. 5
- b) Discuss 802.11 Sub Mac layer protocols. 4
 - c) Name two networks that allow frames to be packed back to back. Why this feature worth having? 3
- Q6 a)** How congestion Control is performed in Frame Relay 4
- b) Which four attributes are required to perform traffic control measurement in frame relay? Describe them in brief. 4
 - c) Distinguish between x.25 and frame relay technologies. 4
- OR**
- Q6 a)** Draw and explain Frame Relay frame format. 4
- b) Explain FRAD in detail. 4
 - c) What limitation of X.25 is over come in Frame relay? 2
 - d) Compare PCV and SVC in frame relay protocol, 2